

School Bus Tracking Software

Complete Module List – Version 1

Purpose: Master list of functional modules required for the School Bus Tracking platform.

#	Module	Platform	Priority
1	Authentication & User Management	Web + Apps	Core
2	School Management	Admin Web	Core
3	Driver Management	Admin Web	Core
4	Bus / Vehicle Management	Admin Web	Core
5	Route Management	Admin Web	Core
6	Stop Management	Admin Web	Core
7	Student Management	Admin Web	Core
8	Parent Management	Admin Web	Core
9	Student Route & Stop Assignment	Admin Web	Core
10	Daily Trip Assignment	Admin Web	Core
11	Driver Mobile App	Android	Core
12	Parent Mobile App	Android	Core
13	Live GPS Tracking	Backend + Apps + Web	Core
14	Real-Time WebSocket	Backend	Core
15	ETA & Distance	Backend + Apps	Core
16	Geofence & Stop Detection	Backend	Core
17	Trip & Journey Management	Web + Apps	Core
18	Push Notifications	Backend + Apps	Core
19	School / Route Announcements	Admin Web + Parent	Core
20	Delay & Breakdown Management	Web + Apps	Core
21	Dashboard	Admin Web	Core
22	Reports & History	Admin Web	Core
23	Audit Logs	Admin Web	Important
24	Settings & Configuration	Admin Web	Important
25	Security & Access Control	All	Core

1. Authentication & User Management

- Login, logout, OTP verification, password/secure staff login where applicable.
- Role-based access control.
- User activation, suspension and profile management.
- School-level data isolation.
- Session/device management.

2. School Management

- School profile and contact details.
- Academic/operational settings.
- School working days and transportation configuration.
- School status and configuration.

3. Driver Management

- Driver registration and profile.
- Driver mobile number and identity details.
- Driver status: Active, Inactive, Suspended.
- Driver trip history.
- Daily assignment support without permanent route binding.

4. Bus / Vehicle Management

- Bus registration and vehicle details.
- Vehicle number, capacity and status.
- Active/inactive/maintenance status.
- Driver and trip history.
- Optional GPS/device information.

5. Route & Stop Management

- Create, edit, activate/deactivate routes.
- Add ordered stops to a route.
- Stop latitude/longitude and geofence radius.
- Pickup/drop timings.
- Route map preview.
- Route direction and trip type support.

6. Student & Parent Management

- Student registration and admission details.
- Class and section.
- Parent/guardian details.
- Multiple children linked to one parent.
- Parent mobile number and OTP identity.
- Student status and transportation status.

7. Student Route & Stop Assignment

- Assign one student to a route.
- Show only stops belonging to the selected route.
- Assign pickup stop and drop stop as applicable.
- Maintain assignment history.
- Allow authorized admin/transport staff to change assignments.

8. Daily Trip Assignment

- Create daily assignment for Date + Driver + Bus + Route + Trip Type.
- Support Morning Pickup and Evening Drop.
- Driver can change between routes/buses on different days.
- Assignment history.
- Trip status lifecycle.

9. Driver Mobile App

- Secure login.
- Today's assignment screen.
- Assigned bus and route details.
- Route stop list and map.
- Start Trip / End Trip.
- Background GPS location service.
- Trip status and operational events.
- Delay, breakdown and emergency reporting.

10. Parent Mobile App

- Mobile number + OTP login.
- Linked children list.
- Switch Child.
- Current trip status.
- Live bus location.
- ETA and distance to child's stop.
- Upcoming / Approaching / Reached / Crossed status.
- Notification history.
- School and route announcements.

11. Live Tracking & Real-Time Engine

- Driver GPS collection.
- Background location handling.
- Go backend location ingestion.
- Redis current-location cache.
- WebSocket live location delivery.
- Real-time parent and admin map updates.
- Last-seen timestamp and stale-location detection.

12. ETA & Geofence

- Calculate distance from bus to assigned stop.
- Calculate estimated arrival time.

- Detect stop arrival using geofence.
- Detect stop crossing after the bus passes the stop.
- Trigger approaching/reached/crossed events.
- Handle GPS accuracy and temporary signal loss.

13. Notifications

- Bus Started from School.
- Bus Approaching Stop.
- Bus Reached Stop.
- Bus Crossed Stop.
- Delay notification.
- Breakdown/emergency notification.
- School reached / trip completed.
- Notification history and delivery status.

14. Announcement System

- School-wide notification to all relevant parents.
- Route-specific notification.
- Title + message + optional attachment.
- Scheduled/manual send support.
- Notification history.

15. Delay & Breakdown Management

- Admin/Transport Manager can record delay reason.
- Traffic, breakdown, road block, weather, driver issue and other reasons.
- Send affected-route notifications.
- Emergency pickup instruction.
- Maintain event history.

16. Admin Dashboard

- Total buses, drivers, students and active routes.
- Live buses currently running.
- Trips started/completed.
- Delayed buses.
- Offline/stale buses.
- Route-wise operational summary.
- Quick access to alerts and active trips.

17. Reports & History

- Trip history.
- Bus journey history.

- Driver trip history.
- Route performance.
- Delay history.
- Student route/stop assignment history.
- Notification history.
- Live location history where retention is enabled.

18. Audit, Settings & Security

- Audit logs for important actions.
- Role and permission configuration.
- School data isolation.
- API authentication and authorization.
- Secure WebSocket authentication.
- Rate limiting and abuse protection.
- Configurable data retention and privacy settings.

19. Recommended Module Dependency Flow

Foundation	Users → Roles → Schools
Transport Master	Drivers → Buses → Routes → Stops
Student Master	Students → Parents → Route → Stop Assignment
Operations	Date → Driver + Bus + Route → Trip
Tracking	Driver GPS → Go → Redis → WebSocket → Parent/Web
Intelligence	Geofence → Stop Event → ETA → Trip Status
Communication	Trip Event → Push Notification
Administration	Dashboard → Reports → Audit Logs

20. Version 1 Scope

Version 1 should focus on reliable route/stop assignment, daily trip operations, background GPS, real-time bus tracking, ETA, geofence-based stop status and targeted notifications. Advanced analytics, attendance automation and additional integrations can be added after the core tracking flow is stable.